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Fault Detection and Diagnosis in dynamic systems using Weightless Neural Networks

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ABSTRACT

This work examines Fault Detection and Diagnosis (FDD) based on Weightless Neural Networks (WNN) with applications in univariate and multivariate dynamic systems. WNN use neurons based on RAM (Random Access Memory) devices. These networks use fast and flexible learning algorithms, which provide accurate and consistent results, without the need for residual generation or network retraining, and therefore they have great potential use for pattern recognition and classification (Ludermir, Carvalho, Braga, de Souto, 1999). The proposed system firstly executes the selection of attributes (in the multivariable case) and does the time series mapping of the data. In the intermediate stage, the WNN performs the detection and diagnosis per class. The network outputs are then passed through a clustering filter in the final stage of the system, if a diagnosis per fault groups is necessary. The system was tested with two case studies: one was an actual application for the temperature monitoring of a sales gas compressor in a natural gas processing unit; and the other one uses simulated data for an industrial plant, known in the literature as "Tennessee Eastman Process". The results show the efficiency of the proposed systems for FDD with classification accuracies of up to 98.78% and 99.47% for the respective applications.

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Abbreviations: ANFIS, Adaptive Neuro-Fuzzy Inference System; ANN, Artificial Neural Net; BA, State of Bahia, Brazil; CW, Counting WiSARD; $Disc_k$, Discriminator k; FCM, Fuzzy C-Means; FDD, Fault Detection and Diagnosis; GA, Genetic Algorithm; HTA, High Temperature Alarm; HHTA, High High Temperature Alarm; kNN, k-Nearest Neighbour; LDA, Linear Discriminant Analysis; MSPCA, Multi-Scale Principal Component Analysis; MTS, Multivariate Time Series; NGPU, Natural Gas Processing Unit; Petrobras, Petróleo Brasileiro S.A.; PIMS, Plant Information Management System; PLS, Partial Least Squares; PW, Pattern WiSARD; QDA, Quadratic Discriminant Analysis; RAM, Random Access Memories; RewPun, REWard/ PUNishment; SES, Simple Exponential Smoothing; SGC, Sale Gas Compressor; SPA, Successive Projections Algorithm; $SPCA_{ms}$, Multiscale PCA Similarity factor; TEP, Tennessee Eastman Process; WiSARD, Wilkes-Stonham-Aleksander Recognition Device; WNN, Weightless Neural Network; $\bar{a}_i$, Average attribute of the t values of the processed series for attribute i; A, C, D, E, Reagents used in TEP; ACC, Accuracy; a_h, Original series attribute in period h; $a_{n,t}$, Attribute n of the input pattern $\boldsymbol{x}_t$; a_t, Input atttribute in instant t; B, Inert gas used in TEP; $B1$, State of broken instrument with readings on top scale (200 °C); $B2$, State of broken instrument with binary oscillation of readings between 0° and 200 °C; $B3$, State of broken instrument with readings reduced for the background scale (0 °C); $B4$, State of partially broken instrument with readings oscillating around the normal operation value; $B5$, State of broken instrument with readings oscillating around values near 0 °C; $B6$, State of broken instrument with readings oscillating around values near 200 °C; BM, Bookmaker informedness; $C[I]$, Memory position accesssed by the vector of inputs I; CN, Condition negatives; CP, Condition positive; C_w, Confidence level of the WiSARD net for the recognition of a pattern w; $d_{a_{i,t}}$, Deviation attribute i in relation to the current input pattern $\boldsymbol{x}_t$; d, Terminal containing required output value; $D1$, $D2$, States with systematic deviations in relation to the normal operation value; DOR, Diagnostic odds ratio; E_h, Value of the series adjusted in period h; E_{h-1}, Value of the series adjusted in period h–1; F, Byproduct generated in TEP; $f(x)$, Function that defines approximately the limits for the classes considered; F_1, F1 score; F_j, Fault pattern type j; FDR, False discovery rate; FN, False negative; FNR, False negative rate; FOR, False omission rate; FP, False positive; FPR, False positive rate; G, H, Final products in TEP; h, Horizon of the processed time window; I, Vector of inputs for the RAM; k, Quantity of discriminators in the WiSARD net; $LR+$, Positive likelihood ratio; $LR-$, Negative likelihood ratio; MCC, Matthews correlation coefficient; MK, Markedness; n, Dimensionality of input vectors for the used application; N, State of normal sensor operation; $n_t(D_{hf})$, Training in the class discriminator with the highest number of patterns; NA, Total number of attributes used in the data set; NC, Number of classes; NPV, Negative predictive value; p, Input RAM node; p, Number of bits of the vector of inputs for the RAM; P, Prevalence; $P_{2,max}$, Second highest score obtained by one of the discriminators of the WiSARD net; P_{max}, Maximum score obtained by one of the discriminators of the WiSARD net; Pun_i, Punishment for the attribute i; PPV, Positive predictive value; q, Phase in which the RAM node is runing: learning or test; r, RAM number of a discriminator; Rec_i, Reward for the attribute i; $R_{i,c}$, Attribute position i in class c; RP_i, Reward and Punishment for the attribute i; SPC, Specificity; T, Total population; TN, True negative; TNR, True negative rate; TP, True positive; TPR, True positive rate; $V_{i,c}$, Value that represents the importance of the attribute i for class c; W, Adjustment coefficient belonging to the interval $0 < W < 1$; w, Input pattern for the weightless neural net; $\boldsymbol{x}$, Input pattern for the system; $\boldsymbol{x}_t$, Input pattern $\boldsymbol{x}$ in instant t; y, Output of the weightless neural net with value equal to a class label; β, Possible values for the conventional bleaching; $\Delta_{a_{i,t}}$, Pertinence degree of the attribute i for the input pattern $\boldsymbol{x}_t$; μ, Arithmetic mean of the t values of the processed time series; σ, Standard deviation of the t values of the processed time series.

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1. Introduction

Both current technological advances and the increasing demand for more productive and reliable industrial processes have resulted in more complex automation with greater data availability. As a result, there is an increasing need for more efficient supervision and control systems, especially for Fault Detection and Diagnosis (FDD) in dynamic environments. In a production system, a fault is an abnormal operating condition caused by factors such as design errors, installation errors, misuse or the effects of natural degradation. The availability of mechanisms for early and reliable detection of faults decreases the risks of malfunction or unscheduled shutdowns of the system. Consequently, it increases equipment reliability and avoids material losses, environmental accidents and harm to workers (Blázquez & Miguel, 2005; Chiang, Russel, & Braatz, 2001; Romano & Kinnaert, 2006; Yang & Liu, 1998).

Considering the *a priori* knowledge used for their development, FDD systems are designed using process models or historical data. Such approaches can be subdivided into quantitative and qualitative methods (Venkatasubramanian, Rengaswamy, & Kavuri, 2003; Venkatasubramanian, Rengaswamy, Yin, & Kavuri, 2003). Systems based on historical data can extract relevant features from the data in order to map the relationships and the existing limits between the considered classes or groups. In this group, some quantitative methods can be highlighted: statistical classifiers (Luo, Wang, & Cui, 2011; Ma, Wong, Jang, & Tseng, 2010; Soares & Galvão, 2010); neural networks (Lau, Ghosh, Hussain, & Hassan, 2013; Leite, Costa, & Gomide, 2012; Sartori, 2012; Zarei, 2012); fuzzy logic (Andrade, 2012; Andrade, Fontes, & Embiruçu, 2011; Lau et al., 2013; Leite, Costa, & Gomide, 2012; Li, Yu, Hilton, & Liu, 2013; Silva, Palhares, & Caminhas, 2012); Principal Component Analysis (PCA) (Barragan, Fontes, & Embiruçu, 2016; Jiang, Yan, & Zhao, 2013; Lau et al., 2013); method of partial least squares (Zhang, Zhou, Qin, & Chai, 2010); wavelet transforms (Barragan et al., 2016). Among the qualitative methods, expert systems (Saravanan, Cholairajan, & Ramachandran, 2009; Wang, Li, & Vrbanek, 2012; Zadeh, 2008) and qualitative trend analysis (Maurya, Rengaswamy, & Venkatasubramanian, 2007) are noteworthy. In the model-based approaches, the actual behavior of the monitored system is compared to the response obtained by a representative model of the process. The result of this supervised comparison is a residual vector used to detect the presence of faults. In this group the following quantitative methods can be highlighted: state and output observers (Chetouani, 2008; Kalman, 1960); parity space and equations (Beckerle, Schaede, Butzek, & Rinderknecht, 2012; Blesa, Jiménez, Rotondo, Nejjari, & Puig, 2014; Zakharov, Tikkala, & Jämsä-Jounela, 2013; Zhong, Song, & Ding, 2015); extended Kalman filter (Kalman, 1960; Patwardhan & Shah, 2006); support vector machine (Zhang, Zhou, Guo, Zou, & Huang, 2012; Deng, Lin, & Chang, 2011; Duan, Xie, Bai, & Wang, 2016; Park, Kwon, Kim, & Baek, 2011); and parameter identification and estimation methods (Johansson, Bask, & Norlander, 2006; Pouliezos, Stavrakakis, & Lefas, 1989). In the qualitative approach, the following methods stand out: fault trees (Nguyen & Lee, 2008; Simões Filho, 2006); qualitative simulation (Berleant, 1991); qualitative process theory (Venkatasubramanian, Rengaswamy, & Kavuri, 2003); and Bayesian networks and other Bayesian reasoning extensions, such as signed directed graphs and evidence theory (Ji, Xia, & Meng, 2015; Luo, Yang, Hu, & Hu, 2012; Xiao, Zhao, Wen, & Wang, 2014). Serdio, Lughofer, Pichler, Buchegger, and Efendic (2014, 2015) have developed approaches that combine historical data, automatic model extraction and residual generation, which seem useful for fault detection and isolation with abrupt and incipient faults.

Many studies have shown the positive advantages of statistically based and artificial intelligence based techniques for solving FDD problems, especially hybrid systems involving fuzzy logic, genetic algorithms and mainly neural networks (Chen & Chen, 2011; Ghate & Dudul, 2010; Mendonça, Sousa, & Sá da Costa, 2009; Niaki & Abbasi, 2005; Sartori, Amaro, Souza Júnior, & Embiruçu, 2012; Sharma, Dewan, & Chatterji, 2015; White & Lakany, 2008; Wu, 2011; Zarei, 2012). The noise tolerance shown by artificial neural networks (Özyurt & Kandel, 1996) and the ability of fuzzy logic to deal with information inaccuracies, ambiguities and uncertainties (Leite et al., 2012) mean that neuro-fuzzy hybrid systems are widely used (Hell, Costa, & Gomide, 2008; Lau et al., 2013). These techniques are well suited to non-linear systems because they do not require explicit mathematical models (Angelov & Yager, 2012; Bartyś, Patton, Syfert, de las Heras, & Quevedo, 2006; Bocaniala & da Costa, 2006; Li et al., 2013; Lo, Fung, & Wong, 2009; Ma et al., 2010; Rigatos & Zhang, 2009). According to Sartori, Amaro, Arduini, Souza Júnior, and Embiruçu (2016), most applications of FDD systems are in power generating and distribution units, pieces of equipment in process industries (reactors, columns, sensors and actuators) and motors and bearings. This study lists neural networks, fuzzy logic, principal component analysis, Kalman filter, support vector machines, genetic algorithms and expert systems as the most commonly used techniques in descending order.

In general, such approaches are well suited to the detection and diagnosis of abrupt faults. In the context of incipient faults in dynamical systems, the FDD problem is more complex and the studies and research found in the literature have yet to find appropriate solutions for problems such as: multivariate problems and a diversity in the number of classes; nonlinear processes with incipient faults present in two or more considered classes; difficulty in obtaining historical data with real-world applications. The main contribution of this paper is to propose a detection and diagnosis system for incipient faults in dynamic systems, without the need to use a mathematical model or residual calculations and with a low false alarm rate. The proposed system is based on Weightless Neural Networks [WNN, an acronysm also used for Wavelet Neural Networks (Lei, He, & Zi, 2011)], initially proposed by Aleksander (1967). WNN are digital models based on Random Access Memory (RAM) devices. Unlike conventional neural models, learning happens in memories inserted into the neuron, in the form of truth tables. Compared to weighted models, WNN have the advantages of a diversity of use of these memories, such as similarity with the conventional digital systems; fast and flexible learning algorithms; accuracy and consistency in the results, no need for generating residuals and network retraining; and above all, great potential for pattern recognition and classification.

The applications of WNN for pattern recognition and classification problems can be found in several areas, including: digit and fingerprint recognition (Bandeira, França, & França, 2009; Conti, Militello, Vitabile, & Sorbello, 2009; Grieco, Lima, De Gregorio, & França, 2010); faces and facial features recognition (Araújo, 2011; Sirlantzis, Howells, & Gherman, 2009; Subhashini & Nagarajan, 2014); robot navigation (McElroy & Howells, 2011; Nurmaini, Hashim, & Jawawi, 2009); data stream clustering (Cardoso, Lima, de Gregório, Gama, & França, 2011); and time series forecasting (De Souza, Freitas, & De Almeida, 2010; Mpofu, 2006). However, no work addressing the problems of FDD with the use of WNN was found in the literature. The papers presented by De Gregorio and Giordano (2014) and Cardoso, De Gregorio, Lima, Gama, and França (2012) approach this context, but do not deal with FDD problems. De Gregorio and Giordano (2014) used WNN for the problem of detecting changes in the vision field of a camera. The proposed system, called CwisarDH, uses a discriminator for each coverage point of the video with the color concept. Cardoso et al. (2012) presented the StreamWiSARD system for flow data grouping with sliding-window. The system consists of WiSARD discriminators as primary units and is able to define high quality clusters, restricted to a small number of microclusters. The absence of

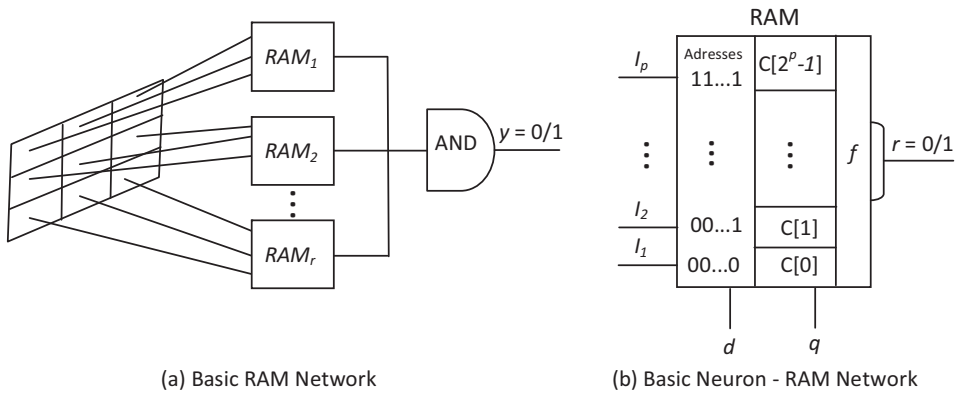


Fig. 1. RAM neuron and network structure.

FDD systems based on WNN, and the potential of these networks for pattern recognition and classification, are the main factors that motivated this work.

In order to validate the developed systems two case studies were considered. The first one uses a univariate database of temperature sensors at a Natural Gas Processing Unit (NGPU) sales gas compressor from Petrobras, located in Pojuca, Bahia, Brazil (Andrade, 2012; Andrade et al., 2011; Sartori, 2012). In the second case study, multivariate data from an industrial plant simulator considered a benchmark in Fault Detection and Diagnosis problems, known as Tennessee Eastman Process (TEP) was used (Downs & Vogel, 1993; Ricker, 1995). In the multivariate application an algorithm of attributes selection per class proposed by Vale, Neto, and Canuto (2010) was used, with modifications proposed in this paper.

This paper is organized as follows: Section 2 presents the fundamentals of WNN; Section 3 describes the FDD-WiSARD (Wilkes-Stonham-Aleksander Recognition Device) system and the method used for training and testing; and in Section 4 the two case studies are used and their respective results are presented. Finally, Section 5 presents concluding remarks.

2. Weightless Neural Networks

Aleksander (1967), one of the pioneers in the field of ANN (Artificial Neural Networks), proposed an entirely digital neuron/node model based on RAM devices. The model gave rise to a class of neurons based on Boolean logic, also known as weightless neurons or RAM-based neurons. Neural networks made up of these neurons are called Weightless Neural Networks (WNN). The learning in these networks consists of changes in the memory contents in truth tables inserted in the RAM. A weightless node learns or recognizes only a part (sub-pattern) of the input pattern presented to the network. The sub-patterns presented as inputs to the RAM correspond to the accessed addresses and they are defined by some input pattern mapping criterion (qualitative or quantitative attribute) binarized to the network, as shown in Figs. 1 and 2. Therefore, the generalization in WNN with RAM nodes only occurs at the network level (Ludermir, Carvalho, Braga, & de Souto, 1999). The network structure shown in Fig. 1(a) refers to the basic weightless neural model defined by Aleksander (1967), named RAM network, which is fully binary and whose neuron is detailed in Fig. 1(b). In WNN, only training and testing steps are executed. The test is performed at the end of training and in many works it is also referred to as validation.

A p input RAM node has 2^p memory locations, addressed by the p bits vector $w=\{w_1, w_2, \ldots, w_p\}$. A binary signal $I=\{I_1, I_2, \ldots, I_p\}$ on the input lines will access only one of the locations, the one for which $w=I$. The terminal q (Fig. 1(b)) defines the phase in which the node is found: learning or testing. If the network is in the learning process, the value of q is equal to 1 and the content accessed by the p input lines is changed to the desired value (0 or 1) provided by terminal d. The content $C[a]$ in all memory locations, where a is the address accessed by the pattern I, is initialized with an 0. In the test phase the bit stored in the memory position $C[a]$ is the output of RAM, $r=C[a]$, defined by the identity function f. If the value found in the memory position $C[a]$ is 0, it means that the last learning assigned an 0 value or that address has not been accessed during the training. Likewise, in RAM, the network output y of the RAM network is also 0 or 1. The network output is equal to 1 only if the outputs of all RAM are equal to 1. Otherwise, the pattern will be included in the class 0 as a pattern not belonging to that application. If more than two classes are required, a discriminator structure (shown in Fig. 2 and described in the following section) is used.

2.1. The WiSARD network

A network composed of RAM neurons may be of several architecture types, although the most used is WiSARD proposed by Aleksander, Thomas, and Bowden (1984). The WiSARD network consists of two or more discriminators and is shown in Fig. 2. Each discriminator is individually trained to represent a single class of patterns. The amount of RAM in each discriminator depends on the mapping model of the input patterns for the network, as can be seen in Fig. 2. Each sub-pattern learned or tested by a RAM is represented by an n-tuple (Bledsoe & Browning, 1959), according to the mapping model used. In the test phase, the same input is presented to all discriminators. Each RAM identifies the memory address pointed to by the corresponding n-tuple and presents the stored value as output, 0 or 1, as in Fig. 2. The output of each discriminator corresponds to the number of RAM with output equal to 1 (sum of numbers 1). The test pattern is recognized by the discriminator which has the greatest sum.

In addition to the scores calculated for each discriminator, the WiSARD network provides a confidence (C_w) level for the recognition performed calculated by:

$$C_w = \frac{P_{\max} - P_{2,\max}}{P_{\max}} \quad (1)$$

where $P_{\max}$ is the maximum score effectively obtained by one of the discriminators and $P_{2,\max}$ is the second highest score. Fig. 3 shows the result of the recognition by a WiSARD with k discriminators. In this illustration, the k discriminator had the maximum score and the discriminator 2 had the second highest score.

When two or more discriminators have the highest score, confidence is null. In this case, if there is a draw with the maximum

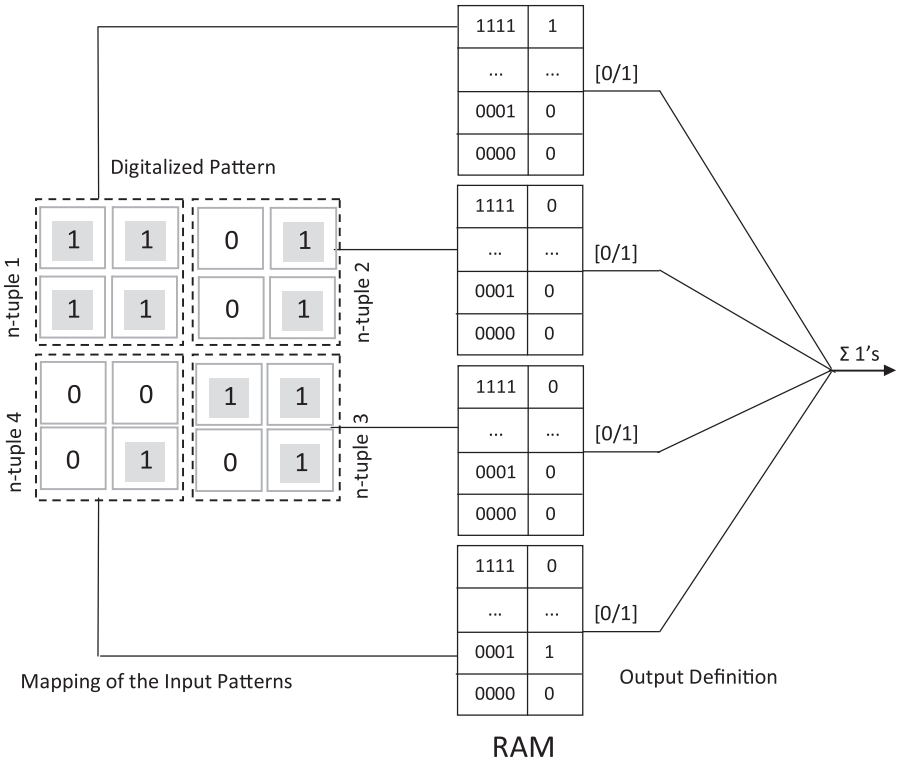


Fig. 2. Schematic representation of a discriminator.

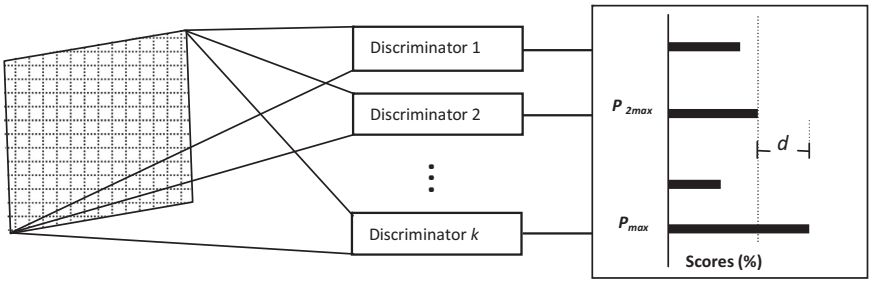


Fig. 3. WiSARD - recognition with k discriminators.

possible score, the randomness criterion is used to define one of the tied-up discriminators as the winner. However, the use of the randomness criterion decreases the accuracy of the network because the tested pattern is probably misclassified. Another problem that occurs in the default (or standard) WiSARD network is the saturation of the memory addresses. Saturation occurs when all or almost all of the RAM addresses are used. Furthermore, the original WiSARD network was originally proposed for the recognition and classification of digital images based on an entirely binary model. This difficulty, however, has been overcome and some papers can be found in the literature in which the standard WiSARD is adapted for applications and procedures with real values, such as the papers presented by Araújo (2011), Cardoso et al. (2011), De Souza et al. (2010), Souza, Nobre, Lima, Silva, Brindeiro, and França (2012) and Subhashini and Nagarajan (2014). Some techniques have also been proposed to overcome the limitations of the randomness criterion and address saturation, which will be discussed in the next section.

2.2. The bleaching technique

The bleaching technique consists of using a threshold value to minimize the use of the randomness criterion and to make a more reliable tie-breaker between the discriminators of a WiSARD network (Carvalho, Carneiro, França, & Lima, 2013). With this technique, the memory addresses in the WiSARD RAM start to store integer values instead of only 0 and 1 values. The integer value stored in a memory address represents the number of times that the address was accessed during the training. In the test phase, the output of a RAM is equal to 1 if the stored integer value is equal to or above the threshold value. Therefore, the threshold value can be implemented in two ways (Carvalho et al., 2013): in its conventional form or in an incremental (or percentage) version.

In conventional bleaching, the same threshold value is used for all the discriminators. Since the threshold is initialized at a zero value, the initial score checking of the RAM behaves as if the technique was not used, *i.e.* if the accessed content is greater than 0, the RAM output is equal to 1. However, if two or more discriminators have the same sum of 1 values, the threshold value increases by one unit. Thus, the contents of the RAM are checked again in order to reduce the number of RAM output equal to 1. This procedure is repeated until a discriminator is elected or until all of them stop scoring, a situation in which the choice is made in a random way.

In the incremental bleaching the threshold value varies according to the amount of learning in each discriminator. This avoids

a class with more training patterns from being privileged because the percentage thresholds have the same representativity and create a balance between the discriminators. Therefore, it is possible to determine the degree of importance of a tested pattern. To do so, conventional bleaching and the number of trainings conducted by each discriminator are used. In this case the output y of a RAM is obtained by Eq. (2):

$$y_j = \begin{cases} 1 & if \quad \frac{C[a]}{nt(D_i)} > \frac{Conventionalbleaching(\beta)}{nt(D_{hf})} \\ 0 & \text{otherwise} \end{cases} \tag{2}$$

where $C[a]$ is the content of the address a accessed by the pattern I, $nt(Di)$ is the number of trainings of the discriminator i containing $C[a]$, with $1 \leq i \leq k$ (k is the number of discriminators, each one with r RAM), β (equal to 0, 1, 2, ...) are the possible values for conventional bleaching and $nt(D_{hf})$ is the number of trainings conducted by the discriminator regarding the class with most patterns.

The bleaching technique may also be used to soften the saturation of memory addresses, as mentioned in the previous section. To do so, a threshold value is taken as reference for gradually erasing less used memory address contents. The erased addresses are related to patterns with little representation or not belonging to the class learned by a given discriminator. In this case, the threshold value is the minimum writing frequency such that a memory content is not erased. The implementation of this procedure consists of storing the content and the last access instant in the RAM memory addresses. In this case, when a memory address is outside the time threshold considered, its content is erased. If all positions of all RAM of a discriminator are erased, this discriminator is discarded because it does not store any information. Therefore, time bleaching is more suitable for unsupervised algorithms, in which the number of discriminators is defined during the learning process (Cardoso et al., 2011; Souza et al., 2012).

3. The proposed wisard-based fault detection and diagnosis system

3.1. System structure

The system proposed in this work performs fault detection as a classification problem in which an input pattern $x=[a_1, a_2, ..., a_n]$, defined as a n-dimensional vector where a_i are the measured variables (attributes), is considered a normal operation or fault state. The diagnosis, on the other hand, consists of the partition of the fault state into subclass labels defined by the different types of faults considered. For this, the attributes $a_1, a_2, ..., a_n$ should contain information that enables the desired classification (Soares & Galvão, 2010). Thus, given an input pattern x and a response variable y, it seeks an approximation of the relationship $y=f(x)$. In this relationship y is a class label valued in the set $\{C_1, C_2, ..., C_k\} \in \aleph^k$, and the function $f(x)$ approximately defines the limits for the considered classes. In a time series context x_t is the input value x at time t, and $a_{1,t}, a_{2,t}, ..., a_{n,t}$ are the attributes used to classify the pattern $x_t=[a_{1,t}, a_{2,t}, ..., a_{n,t}]$ as a normal operating state or a fault state at time t. Regarding its structure, the FDD-WiSARD system contains three distinct and complementary stages, namely (Fig. 4): pre-processing of input patterns; weightless neural network; and clustering filters.

In Fig. 4 $Disc_k$ are the discriminators, with k equal to the number of classes, N is the pattern class with Normal operation and F_j are faulty pattern classes. The initial stage performs the pre-processing of the input patterns (n-dimensional vectors). Each dimension corresponds to a variable (attribute) measured by a sensor or obtained by a simulator. For multivariate applications a modified version of the algorithm of attributes selection per class proposed by Vale et al. (2010) is used, as in Section 3.2.1. The initial stage also performs a time processing procedure, according to

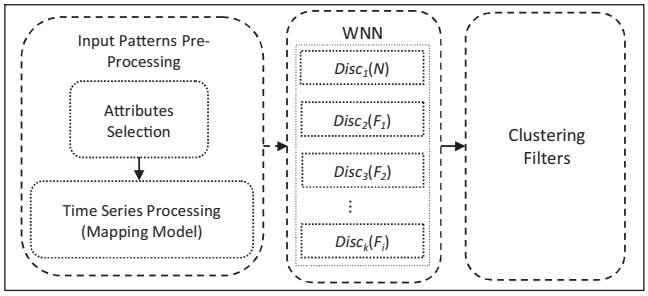


Fig. 4. Structure of the Weightless Neural System: FDD-WiSARD.

the mapping models described in Section 3.2.2. The second (or intermediate) stage corresponds to a WiSARD WNN consisting of a layer with k discriminators individually trained to represent only one class of patterns (Aleksander, 1967). Finally, stage 3 (or final stage) corresponds to clustering filters that reduce the dimensionality of the network output from k possibilities into a desired final classification. These stages of the FDD-WiSARD system are described in more detail in Sections 3.2 to 3.5.

3.2. Pre-processing of input patterns

3.2.1. Selection of attributes

Effectiveness in a pattern classification system consists of carrying out the perfect partition of the state space for the assessed problem. However, this ideal partition is hardly ever obtained, especially in dynamic systems. Notwithstanding, an optimal solution gives results that approximate the ideal situation, considered as the best possible partition of the state space. To do this best possible partition, it is important that the involved classifiers are distinct in the generalizations in order to turn the set of classifiers into a precise and effective system for the recognition and classification of patterns. Thus, one way to obtain a more precise distinction in systems with multi-classifiers, for multivariate problems, is attribute selection (Kuncheva & Whitaker, 2003). The use of different subsets of attributes for the involved classifiers enables the desired distinction in the generalizations. In addition, it is expected that classification errors occur based on established limits for the different classes, according to the sample space distribution (Tsymbal, Cunningham, Pechenizkiy, & Puuronen, 2003).

In this work we used the algorithm of a previous selection of attributes per class called RewPun (Reward/Punishment). This algorithm was proposed by Vale et al. (2010) and is based on reward and punishment to order the importance of the attributes for existing classes. The rational of this model is that there is at least one classifier per class and that the selected attributes for a given class are important only for that class.

The use of the algorithm RewPun in the WiSARD network was possible because the discriminators are individually trained to represent a unique class of patterns. Thus, for each processed pattern the RewPun algorithm selects, in a non-excluding manner, a fixed quantity (N) of attributes per class, using two processes for ranking the importance of the attributes. The first one uses Spearman variance (or correlation) (Guilford, 1950) for ordering and selecting the attributes per class. The second calculates the reward and punishment of the attributes per class based on the rank and selection performed by the first process, which is obtained by means of the parameter RP (Reward/Punishment) defined by Eqs. (3) to (5) (Vale et al., 2010):

$$RP_i = Rew_i - Pun_i \tag{3}$$

$$Rew_i = V_{i,c} + \frac{NA}{NA + R_{i,c}} \tag{4}$$

$$Pun_i = \frac{1}{NC-1} \cdot \sum_{c=1}^{NC, c \neq i} \left(V_{i,c} + \frac{NA}{NA + R_{i,c}} \right) \quad (5)$$

where NC is the number of classes, $V_{i,c}$ is the value that represents the importance of the attribute i for class c, $R_{i,c}$ is the position of the attributes i in class c, and NA is the total number of the attributes used in the dataset (Vale et al., 2010).

When applying the algorithm RewPun to the multivariate case study (Section 4.2), the same attributes were selected for most classes. In order to reach a greater differentiation between the selected attributes for the considered classes, we propose an additional step to the RewPun algorithm, so that the biunivocal relationship between the classes allows a maximum of 50% of common attributes, which is implemented comparing the attributes of a class i, with $i = 1, 2, \ldots, k$, with the attributes of the preceding classes. In this biunivocal comparison, if the number of common attributes is higher than 50%, the first 50% are kept and the other common ones are replaced by less important attributes, according to the second ordering process defined by the RewPun algorithm.

3.2.2. Mapping of input patterns

The mapping models described in this section have the role of transforming the input patterns in RAM access addresses and are referred to as simple mapping or time series mapping. In simple mapping (0), the input patterns are processed individually and presented to the network. In the time series mapping [(1) to (6)], the input patterns are processed in time series with horizon $h > 1$, based on a moving average and sliding window (Hamilton, 1994). Time series mapping transforms a time series into a behavioral pattern to feed the WNN, specifically feeding the neurons (RAM) of the discriminator representative of the behavioral pattern class obtained, when dealing with univariate applications. In comparison, in conventional neural models inputs are multiplied (weighted) by their weights and the result of this weighting feeds the activation function used in the network neurons (Niaki & Abbasi, 2005; Sartori, 2012). In multivariate applications, time mapping is carried out separately for each attribute of input pattern x_t, based on one of the mapping models defined below [(1) to (6)]. In this case, the behavioral pattern is the concatenation of the average attributes found and the training takes place in the discriminator representative of the class of the current input pattern x_t.

0) *Simple mapping.* The current input pattern is processed individually and presented to the network discriminator representative of that pattern class.

1) *Moving average weighted by Hamming distance deviations and average attribute as representative of the processed time series.* In this case weighting is done by Hamming distance (Hamming, 1950) deviations between the current pattern (x_t) attribute and the attributes in previous times. The Hamming distance between two sequences of the same length is the number of positions at which they differ from each other, *i.e.* it corresponds to the number of different bits between two bit vectors of equal size. For the Hamming distances the attributes of the series are represented by 8 or 32 bits, and the mean attribute $\bar{a}_i$ is obtained by Eq. (6):

$$\bar{a}_i = \frac{\sum_{t=1}^{h} a_{i,t} \cdot d_{a_{i,t}}}{\sum_{t=1}^{h} d_{a_{i,t}}} \quad (6)$$

where $d_{a_{i,t}} = |a_{i,t} - a_{i,h}|$ is the deviation of attribute i regarding the current input pattern x_t, with $i = 1, 2, \ldots, n$ and $t = 1, 2, \ldots, h$, and h is the horizon of the processed time window.

2) *Moving average weighted by Hamming distance deviations and the nearest attribute from average attribute as representative of the processed time series.* This model differs from model "1" as it selects $\bar{a}_i$ as the processed time series attribute with less deviation from the average attribute found.

3) *Moving average weighted by the degree of membership and average attribute as representative of the processed time series.* Here the average attribute is weighted by the membership degrees obtained through the Gaussian probability distribution function defined by Eq. (7):

$$f(a_t, \sigma, \mu) = \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma^2}} \cdot e^{-\frac{1}{2} \cdot \left(\frac{a_t - \mu}{\sigma}\right)^2} \quad (7)$$

where a_t is the input attribute and μ and σ are the mean and standard deviation of the t values of the processed time series. The average attribute $\bar{a}_i$ is obtained by Eq. (8):

$$\bar{a}_i = \frac{\sum_{t=1}^{h} a_{i,t} \cdot \Delta_{a_{i,t}}}{\sum_{t=1}^{h} \Delta_{a_{i,t}}} \quad (8)$$

where $\Delta_{a_{i,t}}$ is the membership degree of attribute i relative to input pattern x_t, with $i = 1, 2, \ldots, n$ and $t = 1, 2, \ldots, h$.

4) *Moving average weighted by degree of membership and the nearest attribute to the average attribute as representative of the processed time series.* This model differs from model "3" as it selects $\bar{a}_i$ as the processed time series attribute with less deviation from the average attribute found.

5) *Moving average for the Simple Exponential Smoothing (SES) and average attribute as representative of the processed series.* In this model the weights are adjusted by Eq. (9):

$$E_h = W \cdot a_h + (1 - W) \cdot E_{h-1} \quad (9)$$

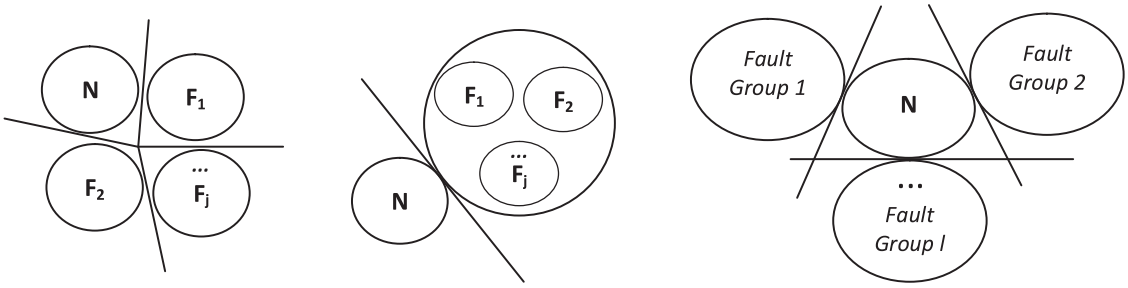
where h is the horizon of the processed time window, a_h is the original series attribute in the period h, E_h is the value of the adjusted series in the period h, $E_{h\text{-}1}$ is the value of the adjusted series in the period h-1 and W is the adjustment coefficient, belonging to the interval $0 < W < 1$. The choice of W is subjective: if interest is in the trend of the series, values near 0 are recommended; if interest is in series prediction or forecasting, values near 1 are recommended.

6) *Moving average by SES and the nearest attribute to the estimated value as representative of the processed series.* This model differs from model "5" as it selects $\bar{a}_i$ as the processed time series attribute with less deviation from the estimated value.

In the time series mapping, the 6 models mentioned take into account the moving average of the attributes of the processed time window. The difference between them is the way this average is calculated. Selecting a suitable mapping model and its parameters for a specific application is an important task. Different systems require different features and different pre-processing, which in our algorithm may be achieved by selecting the most appropriate pre-processing model among those described above and "tuning" its adjustable parameters.

3.3. Structure of the Weightless Neural Network

The WWN consists of a single layer with k discriminators individually trained to represent only one class of patterns. Thus, Normal (N) operation states and Faults (F_1, F_2, ..., F_j) are represented by discriminators D_N, D_{F1}, D_{F2}, ..., D_{Fj}, respectively. As described at the beginning of Section 2, the output of a basic RAM network (Fig. 1(a)) is equal to 1 (class 1) only if the outputs of all RAM are equal to 1. Otherwise, the pattern will be included in another class (0) as a non-class 1 pattern. Thus, it can be stated that a basic RAM network only recognizes patterns that were presented to it in the training phase. However, since more than two categories are required in the applications under study, the structure of the



(a) A classifier for each class. (b) A classifier to detect all faults. (c) A classifier to detect and diagnose a group of faults.

Fig. 5. Detection and diagnostic schemes with the FDD-WiZARD systems.

WiSARD network, described in Section 2.1, was used here. In this network, each discriminator corresponds to a basic RAM network, trained with patterns of a single. In the validation of the WiSARD network, as described in Sections 2.1 and 2.2, the test pattern is presented to all discriminators and classified as belonging to the discriminator class with the highest sum of 1 values as outputs of the RAM (highest number of RAM with output equal to 1), according to the model presented in Fig. 2. Therefore, the criterion "highest", instead of the criterion "all" as in basic RAM, allows discriminators in the WiSARD network to recognize patterns that were not presented to it during the training stage. Then, while the method needs patterns of faulty data, it has some ability to discriminate patterns different from those presented in the training process, especially if the new patterns are similar to known ones. Although not carried out here, training systems without data of a specific type of fault and testing it with data containing this type of fault would be a possible robustness measure of the system regarding fault patterns never seen before in the training stage.

As will be shown later in Section 3.5, two WiSARD network models (Default and Counting) were used for the task of Fault Detection and Diagnosis by the WiSARD-FDD system, as described below. The Default WiSARD (DW), as described in Section 2.1, has a single layer with two or more discriminators individually trained to represent a single class of patterns. In the test phase the same input is presented to all discriminators, thus each RAM node identifies the memory address by the corresponding n-tuple and returns the stored value (0 or 1). The output of the network corresponds to the discriminator with the highest sum of 1 values returned by the RAM nodes. In the case of a draw, the randomness criterion is used to set the winner discriminator. The Counting WiSARD (CW) network has the same structure as the DW, with differences in its functionality. The scalar value stored as memory contents enable the use of bleaching, instead of only 0 or 1, which mitigates the randomness decision in the case of a draw, as described in Section 2.2.

Besides the variation in network and data mapping models, other parameters were evaluated, such as: i) input patterns for Hamming distances with 8 or 32 bits; ii) variation in the time window; iii) input patterns for the network equal to 8, 12, 16 or 32 bits times m (number of selected attributes); iv) number of inputs into the RAM equal to 2, 3, 4, 6 or 8 bits [Fig. 1(b)]; v) number of RAM per discriminator equal to the exact division between items (iii) and (iv); vi) use of simple or incremental bleaching; vii) application dimensionality; viii) attributes selection method (RewPun or modified RewPun); and ix) number of selected attributes (1, 2, ..., n). Items vii, viii and ix refer only to multivariate applications and do not apply to univariate cases.

3.4. Clustering filters

As mentioned earlier, the recognition and classification of a test pattern are performed by one of the k discriminators that make up the WNN. The WiZARD network is equivalent to a multi-classifier structure, capable of detecting and diagnosing faults in parallel and in a single procedure, with a diagnosis per class as shown in Fig. 5(a). However, from the results obtained by the WNN, the clustering filter is optionally used as the final step of the FDD-WiSARD system if a classification per fault group is required, as in the schemes shown in Fig. 5(b) and (c). The clustering filter only groups the discriminated faults in their respective groups. Here two filters are applied: one for the classification in normal state or failure (deviation and broken); and another for classification in normal state or groups of faults [in this case, Deviation ($D1$ and $D2$) or Broken ($B1$ to $B6$) groups]. Therefore, the results at the network level (a more granulated diagnosis) or on the system as a whole can be analysed, similar to the results and analyses presented in Sections 4.1.1 and 4.2.1.

3.5. Method for training and testing the FDD-WiSARD system

Fig. 6 summarizes the method used for Fault Detection and Diagnosis with the FDD-WiSARD system, where n is the number of attributes and h is the horizon of the time window. As mentioned earlier, the step of selecting attributes in the pre-processing of data is performed only for multivariate applications. In time series mapping, six models ("b" to "g") for pre-processing the time series formed for each attribute that makes up the input pattern x_t were tested, as described in Section 3.2.2.

Once the mapping has been carried out, the resulting behavioral pattern is presented to the WNN. In order to assess the importance of the bleaching technique in WiSARD networks, DW and CW models were tested, as described in Section 3.3. Thus, the results of the datasets were evaluated at the levels of the network and the overall system. Since the network has one discriminator per class, the output corresponds to the percentage of successful diagnosis for the normal operating state and for each type of fault state considered. However, other groups may be obtained by applying a specific clustering filter in the final stage of the FDD-WiSARD system. Because fault detection is performed by the WNN at the same time as diagnosis, a grouping of all faults in a single group (a fault detection clustering filter) classifies the patterns as either Normal operating state or Fault state, which in most FDD systems is performed prior to diagnosis.

In the analysis of system performance, the results are presented in terms of quantity and rate of success by pattern class. The results obtained by the FDD systems are presented through a classification matrix with a variety of contents. The criterion for

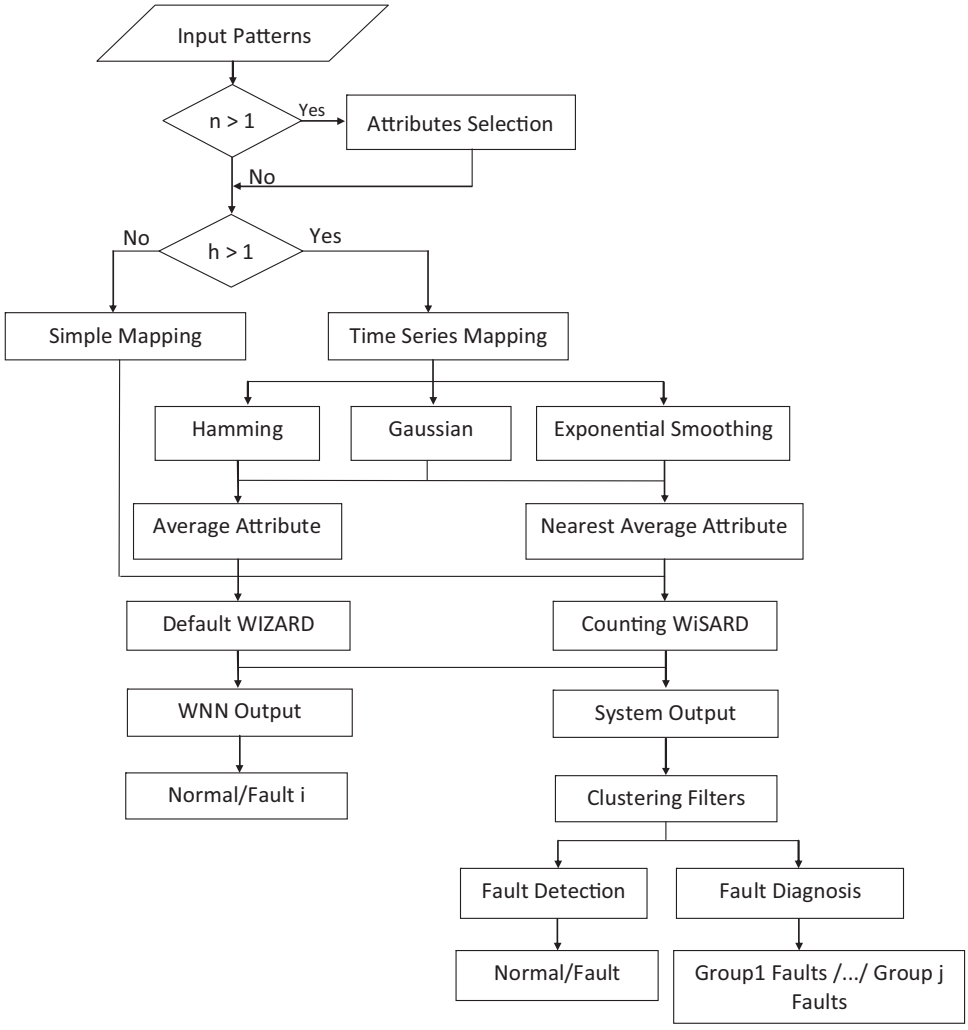


Fig. 6. Method for training and testing the FDD-WiSARD system.

choosing the final system for each case study was the accuracy of the system in the detection and diagnosis of each application. The percentages of successes/errors refer to the due/undue classifications of the data of the test set, based on the established limits for the considered classes. The accuracy of the systems, both for detection and for diagnosis (as shown in Fig. 5 and explained in Section 3.4) is obtained in the system output as a whole (network coupled with the clustering filter) as shown in Tables 5, 6, 12, 13 and 15.

4. Case studies

4.1. Natural gas processing unit

For this case study a univariate database with temperature measurements obtained from an industrial plant was used. Temperature sensors are coupled to a Sales Gas Compressor (SGC) in a Natural Gas Processing Unit (NGPU) at Petrobras, located in the town of Pojuca, Bahia state, Brazil (Andrade, 2012; Andrade et al., 2011; Sartori, 2012). This compressor works with a minimum load of 50% and its functional structure includes an electric motor, a crankcase, five compression cylinders, two intake balloons, two exhaustion balloons and five temperature measuring sensors, one for each cylinder, see Fig. 7. The temperature sensors monitor the safety of the system and detect possible mechanical problems in the compression cylinders. The temperature sensor scales range from 0 °C to 200 °C and measurements are considered normal when their values are close to and within the range of between 80 °C and 100 °C. The temperatures are considered high and very high when they exceed the limits of 140 °C and 160 °C, respectively. The monitoring system is programmed to trigger the HTA (High Temperature Alarm) and the HHTA (High Temperature Alarm) alarms when their respective limits of 140 °C and 160 °C are reached, and triggering the second alarm causes the compressor to trip and the plant to shutdown. In addition to industrial gas

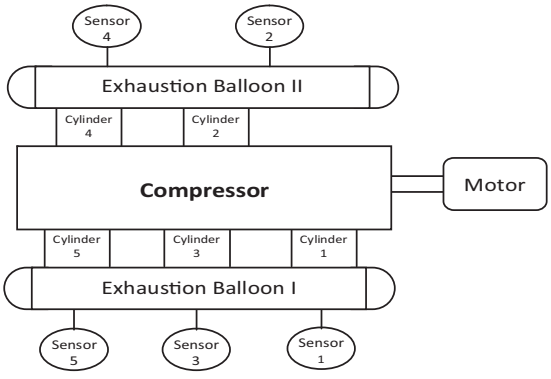


Fig. 7. Schematic representation of the sale gas compressor of the NGPU under study.

conditioning and compression functions for transportation to local companies, the SGC provides kinetic energy to the gas to restore its operating pressure after the losses that occur in the cooling and fractionation steps.

In collaboration with the plant engineers and operators the following steps were taken to build the database: data collection, identification of the fault patterns, and extraction of samples from the identified patterns (Andrade, 2012; Andrade et al., 2011; Sartori, 2012). In the "data collection" step the temperature measurements were recorded by a Plant Information Management System (PIMS). The sampling time was 30 s, the minimum PIMS storage time, and approximately 600,000 temperature measurement points were collected. These measurements correspond to about 208 days, so that almost 7 months of the plant operation were monitored, which highlights the robustness of the method presented here. The step "identification of failure patterns" consisted of the selection and characterization of possible failures, identifying the behavior of the sensors in their different states. The patterns were selected based on the operation history for steady and transient states, not considering cases of plant startup and shutdown. In order to develop and train the neural networks, two datasets, one for training and another for validation of the developed systems, were obtained in the "extraction of samples from the identified patterns" step.

Training and validation datasets were labeled as: "Normal Pattern (N)", to represent the measurements obtained with the normal operation of the sensor and considered correct; or "Deviation 1 ($D1$)" and "Deviation 2 ($D2$)" for the measurement states with deviation; or "Broken 1 to 6 ($B1$, $B2$, $B3$, $B4$, $B5$, $B6$)", for the situation in which the sensor is broken. Faults of type Deviation states occur when Normal states are deviated to values between 100 °C and 130 °C. Although they are regarded as faults, there is no plant shutdown or trigger of alarms. However, faults considered as Broken states pose a high risk and they happen because of high levels of compressor vibration. This vibration at high levels may cause the rupture of the sensor transmission devices and consequently compressor shutdown after HHTA alarm triggering. Although other reasons can cause the same fault states, the vibration of the compressor is considered the main cause of the occurrence of such faults. The conditions for the occurrence of faults are the deterioration of the measuring element and the sensor stem, resulting in events/defects that begin with a shift from the normal condition and cause an offset (systematic deviation) and that may even cause the rupture of the sensor stem (Broken state). Faults of type Broken states are specified as: $B1$, abrupt changes in broken sensor readings to the top of scale (200 °C), with TAHH triggering (Fig. 8(g–h)); $B2$, readings with bottom (0 °C) or top (200 °C) scale values (Fig. 8(i–j)); $B3$, abrupt changes in the readings towards bottom scale (0 °C) (Fig. 8(k–l)); $B4$, variation around the normal operating value (Fig. 8(m–n)), without complete breaking of the stem and with read data transmission to the supervisory system; $B5$, variations with values close to 0 °C, but without reaching the bottom scale (Fig. 8(o–p)); $B6$, variations with values close to 200 °C, but without reaching the top of scale (Fig. 8(q–r)). In order to avoid these undesired states, the developed system seeks to diagnose Normal, Deviation and Broken states of the sensors over time.

4.1.1. Results and analysis - NGPU

As previously mentioned, the system proposed here performs fault detection as a classification problem where an input pattern is considered a normal operation or a failure state. In addition, the diagnosis consists of the partition of the fault state in subclasses with labels defined by the different fault types considered. Because it is a univariate case study, the attributes selection step is not performed, as mentioned in Section 3.1. Thus, data pre-processing results in the implementation of the simple mapping model or of one of the six models of time series mapping described in Section 3.2.2, considering the other parameters mentioned in Section 3.3. Considering the sample time of 30 s, Table 1 shows the number of temperature points and the total operating time of each pattern.

As an illustration, Fig. 8 shows the data sets used in training and testing (validating) the systems for the different states of operation. The validation results correspond to the best performance configuration of the FDD-WiSARD system. Fig. 8(a–b) refers to the training and validation data for the Normal operating state (N), and the others refer to the Deviation ($D1$ and $D2$) and Broken ($B1$ to $B6$) fault states.

Table 2 summarizes the success rates for Fault Detection and Diagnosis obtained by the DW and CW networks for each input pattern mapping model. Considering also the parameters discussed in Section 3.3, Table 2 shows the best performance result in the validation of the FDD systems for each WNN/mapping model pair. Based on all the experiments carried out, Table 3 shows the set of parameters that yields the best performance of the FDD-WiSARD system for each pair WNN/mapping model shown in Table 2.

As seen in Table 2, the network DW performed slightly better in detecting fault state patterns with 4 out of the 7 mapping models used. However, the detection for Normal operation state patterns and fault diagnosis were significantly lower. Because the accuracy of the developed system depends on the detection and diagnosis of faults, the CW network with mapping model 2 (CW-2) perfomed the best for the FDD-WiSARD system. Table 4 details the results obtained by this network.

As already mentioned, the sensor temperature measurements are within a 0 °C to 200 °C range and are considered normal when they are close to and between 80 °C and 100 °C, and they are considered high and very high when they exceed their respective limits of 140 °C (HTA alarm) and 160 °C (HHTA alarm and plant shutdown). Only 3 (0.31%) patterns labeled as Normal operating state were wrongly detected and diagnosed as Broken type faults, 2 of $B3$ type and 1 of $B6$ type. In this case, the improper classifications of Normal patterns would not trigger the trip system, considering that the information is regarded as false, and if the classification were correct the trip system would also not be triggered, because in all three cases the temperature values were below the interlock limit. Therefore these three incorrect classifications would not cause any problems for the system operation. All patterns labeled as Deviation state ($D1$ and $D2$) were correctly detected as a fault state, though 1 pattern $D1$ and 1 $D2$ were improperly diagnosed as a Broken type faults, more specifically $B6$ type. Misclassifications with the Deviation state are less worrying than those related to the Normal pattern because both correct classifications as Deviation and misclassification as Broken would not be considered in the trip logic as they are understood as false temperature measurements.

Since there are no misclassifications of the Normal state to the Deviation or Broken fault states with high temperatures, misclassifications with Broken patterns ($B1$, $B2$, …, $B6$) are the most undesirable, because temperatures above 160 °C indicated by a broken sensor would cause undue plant shutdown. In this case the main concern with the improper trip is economic, whereas not triggering a trip for the Normal and Deviation states with real high temperatures, which does not happen, would cause accidents with possible damage other than economic. As shown in Table 4, 24 Broken patterns were classified in fault detection as Normal operating states: 23 faults of the type $B3$; and 1 of $B5$ type. However, these incorrect classifications would not cause undue plant shutdown, because the temperatures are below 160 °C. In fault diagnosis, 18 Broken patterns were incorrectly classified as Deviation states. From these patterns, 3 failure patterns of $B3$ type (in a total of 13) and 2 of $B4$ type (in a total of 4) showed temperatures above 160 °C, which would cause the improper plant shutdown. However, as these 5 patterns were classified as Deviation states, the

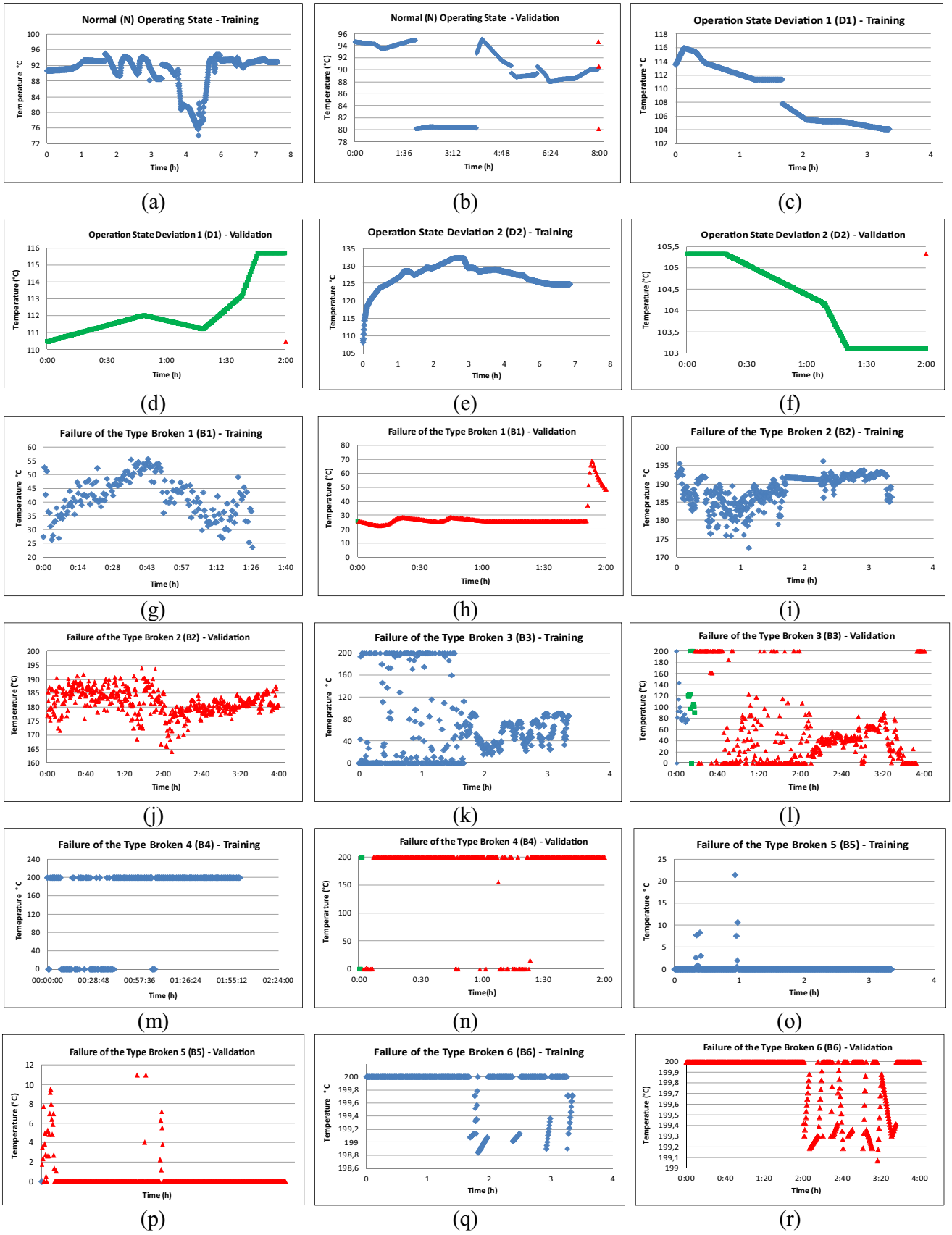


Fig. 8. Data used in the training and testing (validation) of FDD systems for the UPGN case. In validation: ◆ entries with outputs classified as normal; ■ entries with outputs classified as deviation; ▲ entries with outputs classified as broken.

Table 1
FDD-WiSARD system training and test (validation) data samples - NGPU.

Labels/Classes	Training			Test (validation)		
	Time (h)	No. of points	Figure	Time (h)	No. of points	Figure
Normal (*N*)	8.0	954	7(a)	8.0	961	8(b)
Desviation 1(*D*1)	4.0	482	7(c)	2.0	241	8(d)
Desviation 2 (*D*2)	7.0	844	7(e)	2.0	241	8(f)
Broken 1 (*B*1)	1.5	175	7(g)	2.0	241	8(h)
Broken 2 (*B*2)	4.0	482	7(i)	4.0	482	8(j)
Broken 3 (*B*3)	4.0	482	7(k)	4.0	483	8(n)
Broken 4 (*B*4)	2.0	241	7(m)	2.0	241	8(l)
Broken 5 (*B*5)	4.0	482	7(o)	2.0	478	8(p)
Broken 6 (*B*6)	4.0	482	7(q)	2.0	482	8(r)
Total	**38.5**	**4624**		**28.0**	**3850**	

Table 2
Accuracy of Fault Detection and Diagnosis with the FDD-WiSARD system based on pre-processing models for DW and CW networks.

Pre-processing model	Default WiSARD (DW) network				Counting WiSARD (CW) network			
	Fault detection		Fault diagnosis		Fault detection		Fault diagnosis	
	Normal	Fault	Deviation	Broken	Normal	Fault	Deviation	Broken
Simple mapping (0)	50.05	99.45	58.38	98.84	100.00	98.96	100.00	97.93
Hamming/average pattern (1)	21.23	94.05	31.33	86.79	98.44	99.13	91.08	97.42
Hamming/nearest average pattern (2)	49.53	99.30	45.02	97.56	**99.69**	**99.17**	**99.59**	**98.26**
Gaussian/average pattern (3)	22.79	85.19	31.53	56.75	93.34	89.49	65.35	85.64
Gaussian/nearest average pattern (4)	49.22	99.20	60.58	98.67	99.37	99.10	99.17	98.17
SES/average pattern (5)	21.12	91.19	30.29	83.39	98.96	98.44	60.17	96.47
SES/nearest average pattern (6)	50.26	99.20	56.02	98.05	99.69	98.93	99.17	97.34

The bold values signify the best performance system.

Table 3
Parameters that yield the best performance of the FDD-WiSARD system for each pair WNN/mapping model shown in Table 2.

Pre-processing Model	Parameters (see table legend* for i to vi definitions)											
	Default WiSARD (DW) netwok						Counting WiSARD (CW) netwok					
	i	ii	iii	iv	v	vi	i	ii	iii	iv	v	vi
Simple mapping (0)	–	1	12	3	4	–	–	1	12	3	4	1
Hamming/average pattern (1)	8	5	12	3	4	–	32	5	12	3	4	1
Hamming/nearest average pattern (2)	32	35	12	3	4	–	8	5	12	3	4	1
Gaussian/average pattern (3)	–	5	12	3	4	–	–	10	12	3	4	1
Gaussian/nearest average pattern (4)	–	5	12	3	4	–	–	5	12	3	4	1
SES/average pattern (5)	–	10	12	3	4	–	–	5	12	3	4	1
SES/nearest average pattern (6)	–	40	12	3	4	–	–	5	12	3	4	1

* i) standard-size/Hamming-mappings (no. of bits); ii) time-window; iii) network input patterns (no. of bits); iv) RAM inputs (no. of bits); v) no. of RAM per discriminator; vi) simple (1)/incremental (2) bleaching.

Table 4
Fault Detection and Diagnosis of the FDD-WiSARD system with mapping model 2 and counting WiSARD (CW-2) network.

Labels/classes	Fault detection		Tested patters	Fault diagnosis	
	Normal	Fault		Deviation	Broken
N	958	3	961	0	3
*D*1	0	241	241	240	1
*D*2	0	241	241	240	1
*B*1	0	241	241	1	240
*B*2	0	482	482	0	482
*B*3	23	460	483	13	447
*B*4	0	241	241	4	237
*B*5	1	477	478	0	477
*B*6	0	482	482	0	482

Table 5
Probability success (confusion) matrix of the system diagnosis obtained with the counting WiSARD network for mapping model 2 (CW-2).

Classification		Predicted		
		Normal	Deviation	Broken
Real	Normal	**0.9969**	0.0000	0.0031
	Deviation	0.0000	**0.9959**	0.0041
	Broken	0.0100	0.0075	**0.9826**

The bold values signify success probabilities.

right procedure would rule out this information from the trip system, avoiding undue plant shutdowns. Errors occurring with Broken patterns *B*3 are the most justified, because in these cases the stems of the sensors did not completely break, and there is a possibility of data transmission. Thus, the 23 *B*3 patterns classified as Normal state and 10 (from 13) classified as Deviation may have been real (true) readings. Although also unwanted, Broken pattern misclassification at temperatures below 160 °C causes minor impacts, however, they could also be important in decision making.

Table 5 shows a typical confusion matrix (Simon & Simon, 2010), also known as diagnosis probability matrix or error matrix (Stehman, 1997), for the validation dataset used in the FDD-WiSARD system. This representation indicates the probability of the predicted states (possible system outputs) for each real (true) class.

Table 6
Confusion matrix of the system detection obtained with the counting WiSARD network for mapping model 2 (CW-2).

		Predicted condition			
	Total population $T = 3850$	Prediction Positive $PP = 2868$	Prediction Negative $PN = 982$	Prevalence $P = 75.0\%$	
True condition	Condition Positive $CP = 2889$	True Positive $TP = 2865$	False Negative (type II error) $FN = 24$	True Positive Rate $TPR = 99.2\%$	False Negative Rate $FNR = 0.8\%$
	Condition Negative $CN = 961$	False Positive (type I error) $FP = 3$	True Negative $TN = 958$	False Positive Rate $FPR = 0.3\%$	True Negative Rate $TNR = 99.7$
	ACCuracy $ACC = 99.3\%$ F1 score $F_1 = 99.5\%$ Bookmaker Informedness $BM = 98.9\%$	Positive Predictive Value $PPV = 99.9\%$	False Omission Rate $FOR = 2.4\%$	Positive Likelihood Ratio $LR+ = 317.7$	Diagnostic Odds Ratio $DOR = 38120.4$ Matthews Correlation Coeficiente $MCC = 98.2\%$ Markedness $MK = 97.5\%$
		False Discovery Rate $FDR = 0.1\%$	Negative Predictive Value $NPV = 97.6\%$	Negative Likelihood Ratio $LR- = 0.0$	

Success probabilities for the three states are higher than 98%. The Normal state presented the highest probability of success, without any confusion with the Deviation state and with only 3 patterns confused with the Broken state. The Deviation state has a success probability which is 0.1% lower than that of the Normal operating state and a confusion of 2 patterns with the Broken state. The Broken state was the one with the lowest probability of success (98.26%) and a confusion of 24 patterns with the Normal state and 18 with the Deviation state. Based on success percentages for Normal, Deviation and Broken states, the total accuracy of the FDD-WiSARD system is 98.78%.

Considering only fault detection, that is, fault states (condition positive, *CP*), composed of the deviation and broken states, and no-fault states (condition negative, *CN*), conposed of the normal state, Table 6 shows the 2×2 confusion matrix for the validation dataset used in the FDD-WiSARD system accompanied by several indicators derived from it, which provides a more reliable analysis of the performance of the developed system (Fawcett, 2006; Powers, 2011; Ting, 2011):

$$T = CP + CN \tag{10}$$

$$P = 100 \cdot \frac{CP}{T} \tag{11}$$

$$TPR = 100 \cdot \frac{TP}{CP} \tag{12}$$

$$TNR = 100 \cdot \frac{TN}{CN} \tag{13}$$

$$PPV = 100 \cdot \frac{TP}{PP} \tag{14}$$

$$NPV = 100 \cdot \frac{TN}{PN} \tag{15}$$

$$FNR = 100 \cdot \frac{FN}{CP} \tag{16}$$

$$FPR = 100 \cdot \frac{FP}{CN} \tag{17}$$

$$FDR = 100 \cdot \frac{FP}{PP} \tag{18}$$

$$FOR = 100 \cdot \frac{FN}{PN} \tag{19}$$

$$A = 100 \cdot \frac{TP + TN}{T} \tag{20}$$

$$LR+ = \frac{TPR}{FPR} \tag{21}$$

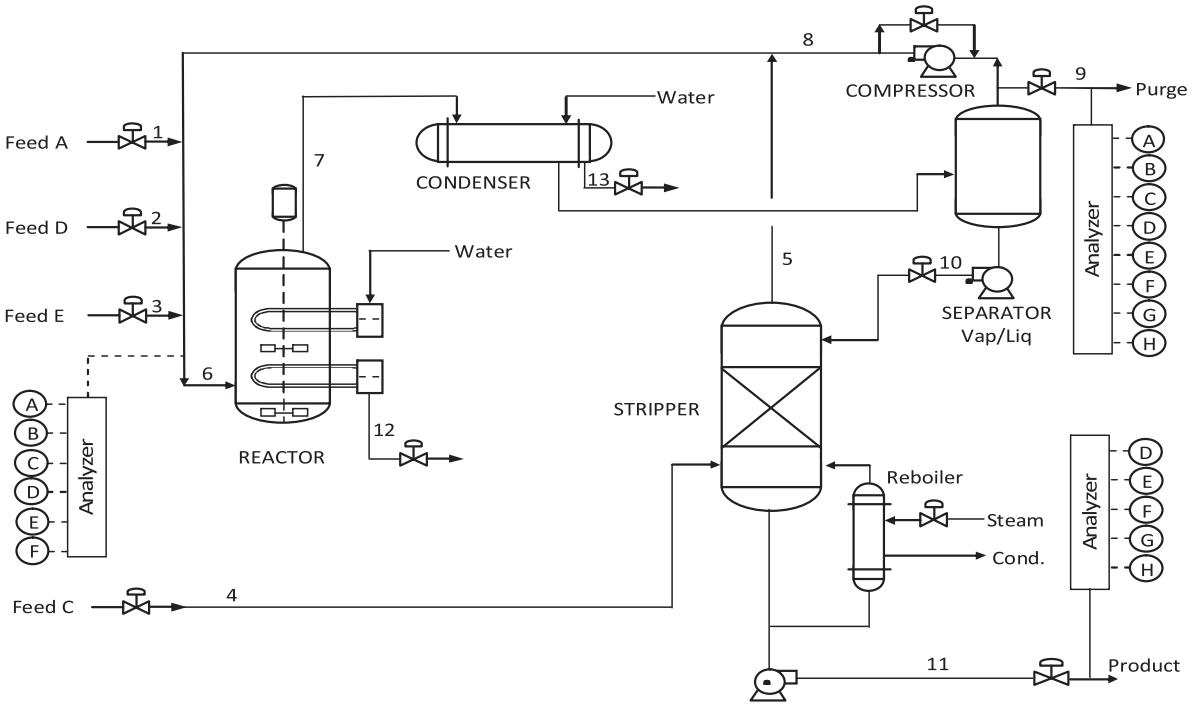


Fig. 9. Schematic diagram of the "Tennessee Eastman Process" (TEP).

$$LR- = \frac{FNR}{TNR} \tag{22}$$

$$DOR = \frac{LR+}{LR-} \tag{23}$$

$$F_1 = 2 \cdot \frac{PPV \cdot TPR}{PPV + TPR} \tag{24}$$

$$MCC = 100 \cdot \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP+FP) \cdot (TP+FN) \cdot (TN+FP) \cdot (TN+FN)}} \tag{25}$$

$$BM = TPR + TNR - 100 \tag{26}$$

$$MK = PPV + NPV - 100 \tag{27}$$

where T is total population, CP is condition positive (the number of real positive cases in the data), CN is condition negatives (the number of real negative cases in the data), P is prevalence, TP is true positive (hit), TN is true negative (correct rejection), FP is false positive (false alarm or type I error), FN is false negative (miss or Type II error), TPR is true positive rate (sensitivity, recall, hit rate or probability of detection), TNR is true negative rate (specificity, SPC), PPV is positive predictive value (precision), NPV is negative predictive value, FNR is false negative rate (miss rate), FPR is false positive rate (fall-out or Probability of False Alarm), FDR is false discovery rate, FOR is false omission rate, ACC is accuracy, $LR+$ is positive likelihood ratio, $LR-$ is negative likelihood ratio, DOR is diagnostic odds ratio, F_1 is F1 score (the harmonic mean of precision and sensitivity), MCC is Matthews correlation coefficient, BM is bookmaker informedness (or simply informedness), and MK is markedness. All indicators demonstrate the successful performance of the developed system, which has high hit rates and low error rates in all classes.

In comparison, the work developed by Andrade et al. (2011) also proposed Fault Detection and Diagnosis systems for this application, using the same database. They used a Fuzzy C-Means (FCM) algorithm to extract the rules used in the pattern classification system based on interval-type fuzzy logic. The results presented by the FDD-WiSARD system (Table 5) and those obtained by Andrade et al. (2011), considering the detection and fault diagnosis for Normal/Deviation/Broken operating states, are respectively: 0.9969/0.9959/0.9826; and 0.9730/0.9970/0.9900. The total accuracy of the Andrade et al. (2011) system was 98.66%. Therefore, the results obtained here are comparable or even better than those obtained with a non-WNN method.

In summary, the final FDD-WiSARD system for Fault Detection and Diagnosis in the sales gas compressor temperature sensors of the NGPU under study has the following configuration: "Initial Stage" - mapping model 2, input patterns to the Hamming distances with 8 bits, time window horizon equal to 5; "Intermediate Stage" - counting WiSARD weightless neural network, input patterns to the network equal to 12 bits, RAM input subpatterns with 3 bits, RAM number for discriminators equal to 4 (12/3) and simple bleaching usage; "Final Stage" - clustering filter for the detection of Normal and Fault states and clustering filter for the diagnosis of Deviation and Broken states.

4.2. The "Tennessee Eastman Process"

The "Tennessee Eastman Process" (TEP) is a model of a chemical process considered a reference in studies of process control and in particular in the field of Fault Detection and Diagnosis (Barragan et al., 2016). The TEP was originally proposed by Downs and Vogel (1993) as an open loop industrial plant with challenges for process control. Subsequently, a number of studies with different control techniques have been carried out, and here the closed loop plant with the control system proposed by Ricker (1995) is considered. Ricker (1995) also presents six different steady states for TEP and one of them was used here in the simulation for generating data for training the FDD-WiSARD system.

As shown in Fig. 9, the TEP comprises an arrangement reactor/separator/recycle with five main operation units: reactor; product condenser; vapor-liquid separator; recycle compressor; and product separation column. The studied plant produces G and H

products from reagents *A, C, D* and *E*, also generating a byproduct *F*, according to the reactions represented by Eqs. (28) to (31). In addition to the reactants and products a *B* inert gas is also used which is introduced into the reagent feed streams.

$$A(g) + C(g) + D(g) \rightarrow G(\text{liq}), Product1 \quad (28)$$

$$A(g) + C(g) + E(g) \rightarrow H(\text{liq}), Product2 \quad (29)$$

$$A(g) + E(g) \rightarrow F(\text{liq}), \text{Subproduct} \quad (30)$$

$$3 \cdot D(g) \rightarrow 2 \cdot F(\text{liq}), \text{Subproduct} \quad (31)$$

Reagents *A, D* and *E* are added to the reactor and reagent *C* with inert *B* feed the stripper separation column. The reactor output passes through the condenser where there is a partial liquefaction by cooling. The condenser output goes to the vapor-liquid separator where a fraction of the gas is purged to avoid the accumulation of inert and by-product in the process, and the other fraction is sent to the centrifugal compressor and recycled to the reactor. The liquid stream of the separator goes to the stripper where reagents are removed from the top and sent to the compressor for recycling to the reactor and the bottom liquid (stream 11) consists essentially of *G* and *H*, the final products of the process.

The TEP model has 12 manipulated variables and 41 measured variables. Among the measured variables, 22 are sampled every 3 min and correspond to flow, temperature and pressure measurements, as shown in Table 7. The other variables are measured at higher sampling rates, and consist of analyzers for the reactor feed, purge and final product. Here 13 different fault conditions in closed loop were considered (Ricker, 1995), described in Table 8. Oscillatory changes consist of random and persistent fluctuations in the respective variable.

4.2.1. Results and analysis - TEP

The data for training and validation of the FDD systems used in TEP were generated using the simulator proposed by Ricker (1995) with the values of the operating mode case 1 proposed by the author. Five conditions were considered, 1 with the original values, which were used for training the networks, and another 4 which were obtained with variations of ±3% and ±5% on the variable production rate and used in the validation of the systems. Ricker (1995) considers this controlled variable as the most relevant in the composition of the final products *G* and *H* and variations in a range of ±5% are considered acceptable. For each condition 14 simulations of 1000 sampling periods of 3 min were performed, one simulating normal operation and the other related to the 13 types of faults. In order to mitigate the effects of initial conditions the first 20 periods of each simulation were discarded so in fact 980 points were used, as shown in Table 9.

By way of illustration, Fig. 10 shows the set of training data related to Normal and Fault 4 operating states, after selection of variables (attributes) with the modified RewPun algorithm and related to the best performance configuration with the FDD-WiSARD system. Fig. 10(a–d) presents variables common to Normal and Fault 4 states, Fig. 10(e–g) shows the other variables related to the Normal operating state and Fig. 10(h–j) complements the variables selected for the Fault 4 state.

The results shown in Table 10 refer to the success rates (accuracy) in the diagnosis by the WNN with the prior use of the modified RewPun attribute selection algorithm. The percentage in each cell in Table 10 corresponds to the best performance for a given configuration (mapping model *versus* network *versus* variation in the production rate), with the other parameters described in Section 3.3. Table 11 shows the set of parameters

Table 7
TEP measured variables.

Variable	Description	t (min)	Unit
y_1	Feeding flow for reagent *A*	3	Kscmh
y_2	Feeding flow for reagent *D*	3	kg/h
y_3	Feeding flow for reagent *E*	3	kg/h
y_4	Feeding flow for reagents *A* and *C*	3	Kscmh
y_5	Compressor recycle flow	3	Kscmh
y_6	Reactor feed flow	3	Kscmh
y_7	Reactor pressure	3	kPa
y_8	Reactor level	3	%
y_9	Reactor temperature	3	°C
y_{10}	Purge flow	3	Kscmh
y_{11}	Product separator temperature	3	°C
y_{12}	Product separator level	3	%
y_{13}	Product separator pressure	3	kPa
y_{14}	Separator outlet liquid flow	3	m^3/h
y_{15}	Stripper level	3	%
y_{16}	Stripper pressure	3	kPa
y_{17}	Stripper outlet liquid flow	3	m^3/h
y_{18}	Stripper temperature	3	°C
y_{19}	Stripper steam flow	3	kg/h
y_{20}	Compressor work	3	kW
y_{21}	Outlet reactor cooling water temperature	3	°C
y_{22}	Outlet condenser cooling water temperature	3	°C
y_{23}	Composition of *A* in the reactor feed	6	mol%
y_{24}	Composition of *B* in the reactor feed	6	mol%
y_{25}	Composition of *C* in the reactor feed	6	mol%
y_{26}	Composition of *D* in the reactor feed	6	mol%
y_{27}	Composition of *E* in the reactor feed	6	mol%
y_{28}	Composition of *F* in the reactor feed	6	mol%
y_{29}	Composition of *A* in the purge gas	6	mol%
y_{30}	Composition of *B* in the purge gas	6	mol%
y_{31}	Composition of *C* in the purge gas	6	mol%
y_{32}	Composition of *D* in the purge gas	6	mol%
y_{33}	Composition of *E* in the purge gas	6	mol%
y_{34}	Composition of *F* in the purge gas	6	mol%
y_{35}	Composition of *G* in the purge gas	6	mol%
y_{36}	Composition of *H* in the purge gas	6	mol%
y_{37}	Composition of *D* in the stripper	15	mol%
y_{38}	Composition of *E* in the stripper	15	mol%
y_{39}	Composition of *F* in the stripper	15	mol%
y_{40}	Composition of *G* in the stripper	15	mol%
y_{41}	Composition of *H* in the stripper	15	mol%

Table 8
TEP simulated faults.

Fault	Description	Change
1	*A*/*C* feed ratio, keeping *B* constant	Step
2	*B* composition, keeping *A*/*C* ratio constant	Step
3	*D* feeding temperature	Step
4	Temperature of reactor cooling water	Step
5	Temperature of condenser cooling water	Step
6	Load (valve) of feeding *A*	Step
7	Load (valve) of feeding *C*	Oscillatory
8	Feed compositions of *A, B* and *C*	Oscillatory
9	*A* feeding temperature	Oscillatory
10	*C* feeding temperature	Oscillatory
11	Temperature of reactor cooling water	Oscillatory
12	Temperature of condenser cooling water	Oscillatory
13	Variation in reaction kinetics (speed, components and parameters)	Oscillatory

Table 9
FDD-WiSARD system training and test (validation) data samples - TEP.

Labels/classes	Training		Test (Validation)**	
	Time (h)	No. of points*	Time (h)	No. of points*
Normal (N)	49	980	49	980
Fault 1	49	980	49	980
…	…	…	…	…
Fault 13	49	980	49	980
Total	643	12,861	643	12,856

* Relative to each input variable;
** Variation of +3% on the production rate.

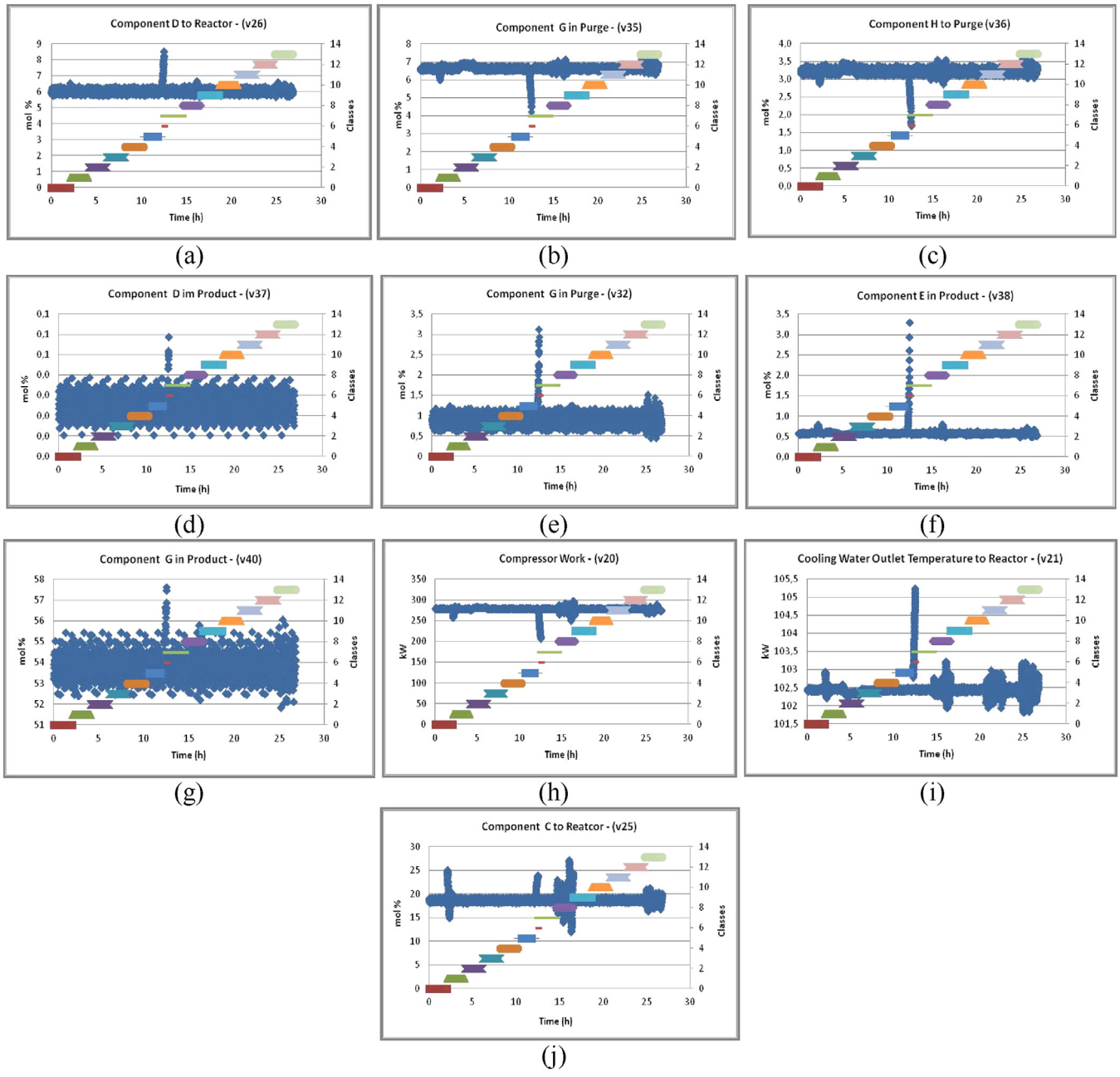


Fig. 10. Data used in the training of FDD systems for TEP - normal and fault 4 operational states.

which performed the best in the FDD-WiSARD system for each pair WNN/mapping model shown in Table 10 for the whole set of TEP simulations available.

Based on the data in Table 10 the following considerations can be made:

a) The CW network performed better than the DW, with significant differences. The low performance in the DW network was because it only saves the last learning in the accessed memory addresses. Furthermore, the use of the randomness criterion for tie-breaking discriminators with 1 summation value greater than 0 affect the performance of the network negatively;
b) The performance of the FDD systems is inversely proportional to the percentage change in the production rate, which justifies the recommendation given by Ricker (1995) not to use variations outside the interval ±5%;
c) The mapping model 0 with CW network (CW-0) was the best performing configuration for diagnosing the best performance equal to 99.47% (+3% variation in the production rate) and the worst one equal to 98.17% (−5% variation in the production rate), with an average success rate equal to 98.79%. Although the mapping model 0 does not use the time series window in the pre-processing of the data, the dynamic behavior of the network is due to the increments in the scalar values stored in the memory addresses. The stored values, in the same addresses or in nearby addresses, record similar behaviors found in the historical data regarding the problem.

The classification matrix shown in Table 12 details the diagnosis of CW-0 configuration (with a variation of +3% in the production rate) per class obtained at the network output with the use of the modified RewPun attribute selection algorithm in the data preprocessing step. In comparison, the classification matrix presented in

Table 10
WNN success rates in percentages (accuracy) for TEP.

Pre-processing Model	Default WiSARD (DW)					Counting WiSARD (CW)				
	−5	−3	+3	+5	Average	−5	−3	+3	+5	Average
Simple mapping (0)	97.35	99.24	99.24	98.15	98.50	98.17	99.32	**99.47**	98.20	98.79
Hamming/average pattern (1)	79.43	95.56	91.33	79.77	86.52	93.65	98.38	98.47	94.89	96.35
Hamming/nearest average pattern (2)	79.68	91.03	83.72	80.47	83.73	92.97	97.57	98.41	93.65	95.65
Gaussian/average pattern (3)	54.27	96.20	94.74	57.87	75.77	90.56	99.07	99.08	93.39	95.53
Gaussian/nearest average pattern (4)	79.59	90.30	89.53	80.37	84.95	93.18	97.48	98.24	93.34	95.56
SES/average pattern (5)	84.20	90.14	93.12	81.03	87.12	94.86	98.70	98.92	94.78	96.82
SES/nearest average pattern (6)	87.02	98.51	97.00	88.36	92.72	94.88	98.21	99.12	96.66	97.22

The bold value signify the best performance system.

Table 11
Parameters that provide the best performance of the FDD-WiSARD system for each pair WNN/mapping model shown in Table 10.

Pre-processing Model	Parameters (see Table legend* for i to vii definitions)													
	Default WiSARD (DW) network							Counting WiSARD (CW) network						
	i	ii	iii	iv	v	vi	vii	i	ii	iii	iv	v	vi	vii
Simple mapping (0)	–	1	84	3	28	–	7	–	1	84	3	28	2	7
Hamming/average pattern (1)	8	5	84	3	28	–	7	8	5	84	3	28	2	7
Hamming/nearest average pattern (2)	8	10	84	3	28	–	7	8	5	84	3	28	1	7
Gaussian/average pattern (3)	–	5	96	3	32	–	8	–	5	84	3	28	1	7
Gaussian/nearest average pattern (4)	–	5	96	3	32	–	8	–	5	84	3	28	1	7
SES/average pattern (5)	–	5	84	3	28	–	7	–	5	84	3	28	2	7
SES/nearest average pattern (6)	–	5	96	3	32	–	8	–	5	84	3	28	2	7

* i) standard size/Hamming-mappings (no. of bits); ii) time window; iii) network input patterns (no. of bits); iv) RAM inputs (no. of bits); v) no. of RAM per discriminator; vi) simple (1)/incremental (2) bleaching; and vii) no. of selected attributes.

Table 12
Classification (confusion) matrix for the counting WiSARD (CW-0) network with a +3% variation in production rate: modified RewPun attributes selection algorithm.

Labels/classes (Real)	(WNN-CW-0 predicted outputs)														Accuracy (%)
	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	
C0	980														100.00
C1		970					10								98.98
C2			980												100.00
C3				975					5						99.49
C4					957	23									97.65
C5						980									100.00
C6							115							1	99.14
C7								980							100.00
C8									980						100.00
C9		4								975				1	99.49
C10											980				100.00
C11												980			100.00
C12		7				4			12				957		9765
C13							1							979	99.90
Outputs	980	981	980	975	957	1007	126	980	997	975	980	980	957	981	**99.47**

Table 13 details the same diagnosis, but with the use of the original RewPun attribute selection algorithm.

Comparing the classification matrixes in Tables 12 and 13, a superior performance of the modified RewPun algorithm can be seen when compared to the original algorithm. The classification confusion with the original RewPun occurred because the selected attributes are the same for most classes, as shown in Table 14(a). To minimize this, as already mentioned in Section 3.2.1, one step was added at the end of the RewPun algorithm so that the biunivocal relationship between classes allows a maximum 50% of common attributes, as shown in Table 14(b).

Considering only fault detection, regardless of diagnosis, Table 15 shows the 2×2 confusion matrix for the validation dataset used in the FDD-WiSARD system accompanied by its indicators. As expected, since the accuracy is 100%, all the indicators perform perfectly in the developed system.

Soares and Galvão (2010) examined fault detection in TEP using classifiers for groups of faults with similar characteristics. For the definition of these fault groups, a non-supervised learning algorithm was used, and the best performance result was the 4 groups shown at the end of Table 16. These authors compared classifiers obtained through parametric methods (Linear Discriminant Analysis, LDA, with Successive Projections Algorithm, SPA, and with Genetic Algorithm, GA; and Quadratic Discriminant Analysis, QDA, with SPA) and nonparametric ones (k-Nearest Neighbour, kNN; and Partial Least Squares, PLS). The best performing result was obtained with SPA-ADQ classifiers, as shown in Table 16. In addition, Table 16 shows a summary of the main quality indicators for the system developed here with the CW-0 configuration and a +3% variation in the production rate of G and H, where a better performance can be seen compared with the Soares and Galvão (2010) methods.

Table 13
Classification (confusion) matrix for the counting WiSARD (CW-0) network with variation of +3% in production rate: original RewPun attributes selection algorithm.

Labels/classes (Real)	(WNN-CW-0 predicted outputs)														Accuracy (%)
	C0	C1	C2	C3	C4	C5	C6	C7	C8	C9	C10	C11	C12	C13	
C0	10			2	37	20		94		87	483	183	64		1.02
C1		970					10								98.98
C2			980												100.00
C3	14			3	35	22		93		80	480	189	64		0.31
C4	12				46	13		90		75	475	199	70		4.69
C5	8			3	36	23		85		87	491	179	68		2.35
C6							115							1	99.14
C7	4			2	23	27		86		82	491	186	76	3	8.78
C8									980						100.00
C9	7			4	34	16		93		79	511	172	62	2	8.06
C10	4			2	21	5		95		82	576	140	55		58.78
C11	9				41	24		110		70	457	206	63		21.02
C12	11			2	46	21		95		78	453	198	76		7.76
C13							1							979	99.90
Outputs	79	970	980	18	319	171	126	841	980	720	4417	1652	598	985	**39.90**

Table 14
Selected attributes per class by the original and modified RewPun algorithms for CW-0 with +3% variation in the production rate.

Classes	Original RewPun	Modified RewPun
0	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 38, 40, 32
1	4, 2, 5, 3, 1, 6, 7	4, 2, 5, 3, 1, 6, 7
2	38, 40, 31, 34, 27, 36, 23	38, 40, 31, 34, 27, 36, 23
3	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 39, 34, 27
4	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 20, 25, 21
5	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 22, 31, 18
6	2, 1, 3, 7, 9, 8, 11	2, 1, 3, 7, 9, 8, 11
7	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 23, 28, 19
8	5, 0, 4, 37, 35, 38, 6	5, 0, 4, 37, 35, 38, 6
9	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 24, 33, 30
10	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 21, 25, 22
11	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 22, 25, 31
12	37, 36, 35, 26, 38, 40, 32	37, 36, 35, 26, 29, 17, 0
13	16, 17, 15, 21, 14, 18, 34	16, 17, 15, 21, 14, 18, 34
	(a) Original RewPun	(b) Modified RewPun

The results presented by Lau et al. (2013) can also be compared with the results obtained here for this TEP case study. They proposed a fault diagnosis system based on Multi-Scale Principal Component Analysis (MSPCA) and Adaptive Neuro-Fuzzy Inference System (ANFIS), referenced as MSPCA-ANFIS and composed of two main steps. The first step performs feature extraction using MSPCA and the second performs pattern classification with ANFIS. The ANFIS in the MSPCA-ANFIS model is a multiclassifier system with purposes similar to the WiSARD here. In the MSPCA-ANFIS, each ANFIS classifier performs the detection and diagnosis of a single type of fault based on a predefined threshold, a function similar to the technique of bleaching in the CW network. The results obtained with the model MSPCA-ANFIS presented a succesful classification percentage of 99.52% for normal operation patterns and a mean percentage of 90.7% for the fault states of the 21 faults analyzed in the TEP process, excluding faults 3, 9 and 15. In the model proposed here, these percentages were 100%, as shown in Table 16.

The Multivariate Time Series (MTS) pattern recognition system presented by Barragan et al. (2016) is based on a combination of wavelet features, PCA similarity metrics and fuzzy clustering, where a Multiscale PCA Similarity factor ($SPCA_{ms}$) is used to determine the distances of the objects (MTS) to be recognized. This factor is obtained from the signal analysis of some TEP variables using the wavelet transform. The database was obtained from the TEP simulator, proposed by Ricker (1995), and presents some specific characteristics, such as: it contains data regarding the normal state of operation and fault 1 state; it uses 4 of the 41 TEP variables; fault events and normal operation data were collected with different changes in operational modes (steady states). The $SPCA_{ms}$ metrics had a success classification rate of 95% for clusters 1 (fault) and 2 (normal), and with traditional SPCA these rates were 70% and 80%, respectively.

Therefore, the results obtained here are comparable or better than those obtained with non-WNN methods. From the results and comparisons above, the final structure of the FDD-WiSARD system for the TEP model has the following configuration: “Initial Stage” - modified RewPun attributes selection algorithm, simple mapping model 0; “Intermediate Stage” - counting WiSARD WNN (CW), input patterns to the network equal to 84 bits (12•7), RAM input subpatterns with 3 bits, RAM number per discriminator equal to 28 (84/3), use of incremental bleaching; “Final Stage” - clustering filter for detecting Normal and Fault states and clustering filter for fault groups diagnosis, according to those defined by Soares and Galvão (2010) and shown in Table 16. The execution time of training and testing the system for this configuration was 3 s. The training and validation data were simulated based on the operating mode case 1 proposed by Ricker (1995) with a +3% variation in the production rate for the validation set.

5. Conclusions

This work addresses the problem of Fault Detection and Diagnosis in dynamic systems with the use of weightless neural networks. The DW network was used as reference for the development of the proposed systems and these systems were tested and validated with univariate data, representing a real problem (historical data from a NGPU, focusing on the temperature monitoring of its sales gas compressor), and with simulated multivariate data from the so-called TEP benchmark problem. With the results obtained for the two case studies it was found that the WNN presented a good solution for the Fault Detection and Diagnosis problem in univariate and multivariate dynamic systems. This was possible given the contribution of several important elements: selection of attributes (variables) per class as an important contribution to the pre-processing of multivariate data; data pre-processing through simple and time series mapping models, with sliding windows based on Hamming distances, Gaussian probability distribution function and SES; the use of the counting approach in pattern recognition and classification performed by the CW network; the use of the bleaching (simple and incremental) technique to minimize the use of randomness in the outputs of the WNN; and the grouping (clustering) filters used to obtain diagnosis per fault groups, since the WNN developed perform Fault Detection and

Table 15
Confusion matrix of the counting WiSARD (CW-0) network detection with a +3% variation in production rate: modified RewPun attributes selection algorithm.

		Predicted condition			
	Total population $T = 12856$	Prediction Positive $PP = 11876$	Prediction Negative $PN = 980$	Prevalence $P = 92.4\%$	
True condition	Condition Positive $CP = 11876$	True Positive $TP = 11876$	False Negative (type II error) $FN = 0$	True Positive Rate $TPR = 100.0\%$	False Negative Rate $FNR = 0.0\%$
	Condition Negative $CN = 980$	False Positive (type I error) $FP = 0$	True Negative $TN = 980$	False Positive Rate $FPR = 0.0\%$	True Negative Rate $TNR = 100.0$
	ACCuracy $ACC = 100.0\%$ F1 score $F_1 = 100.0\%$ Bookmaker Informedness $BM = 100.0\%$	Positive Predictive Value $PPV = 100.0\%$	False Omission Rate $FOR = 0.0\%$	Positive Likelihood Ratio $LR+ = \infty$	Diagnostic Odds Ratio $DOR = \infty$ Matthews Correlation Coeficiente $MCC = 100.0\%$ Markedness $MK = 100.0\%$
		False Discovery Rate $FDR = 0.0\%$	Negative Predictive Value $NPV = 100.0\%$	Negative Likelihood Ratio $LR- = 0.0$	

Table 16
Quality indicators in the validation of the FDD-WiSARD system (CW-0 with +3% variation in the production rate) in comparison with Soares and Galvão (2010).

Observed items	Original RewPun	Modified RewPun	Soares and Galvão (2010)				
			Parametric			Nonparametric	
			SPA-ADQ	SPA-ADL	GA-ADL	kNN	PLS
Tested patterns	12856	12856	–	–	–	–	–
Success - WNN (no.)	5129	12788	–	–	–	–	–
Success - WNN (%)	39.90%	**99.47%**	–	–	–	–	–
Filter 1: Fault detection (average)	99.42%	100.00%	–	–	–	–	–
Normal operation state	1.02%	100.00%	–	–	–	–	–
Fault operation state	99.42%	100.00%	–	–	–	–	–
Filter 2: Diagnosis per fault groups (Soares & Galvão, 2010)	56.83%	99.20%	93.20%	86.00%	75.12%	72.28%	78.32%
Group 1 (faults: 7, 8, 9 and 10)	78.44%	99.87%	94.41%	85.00%	76.00%	71.70%	79.30%
Group 2 (faults: 1 and 4)	51.84%	98.32%	97.80%	90.70%	76.30%	72.20%	77.30%
Group 3 (faults: 2, 3, 5, 6 and 11)	43.21%	99.85%	94.90%	84.80%	76.20%	74.60%	80.80%
Group 4 (faults: 12 and 13)	53.83%	98.78%	86.00%	82.80%	72.00%	70.60%	75.90%

The bold value signify the best performance system.

Diagnosis per class (individual faults) simultaneously. The results show the success of the developed Fault Detection and Diagnosis systems, with success rates of up to 98.78% and 99.47% for the respective NGPU and TEP applications, with very few confused patterns and very good fault detection indicators as shown in Tables 5, 6, 12, 13 and 15.

The time series mapping model using Hamming distances stands out in the UPGN case study and the simple mapping model in the TEP case study. With the definition of the simple mapping model for TEP, and given the complexity of this application, the robustness of the WNN for pattern recognition and classification is shown in this work. The simple mapping model presents the individual patterns for the neural network. Thus, the recurrence of past contributions occurs only with the WNN and, more specifically, in the mode of storage of the memory contents in the RAM, in the fidelity to the method of input pattern presentation for the network and in the bleaching technique used for the definition of the discriminator representative of the test pattern.

In the WiSARD network each discriminator functions as a classifier of a single class, which demonstrates that Fault Detection and

Diagnosis are carried out simultaneously, thus increasing the computational efficiency and the flexibility of the structure for Fault Detection and Diagnosis. This procedure differs from most of the models proposed for the FDD problem, where a structure is used for fault detection and another structure is used to diagnose the detected faults. In addition, in WNN training and test (validation) are performed without the need to retrain the model, thus giving consistency to the results obtained.

Finally, the FDD-WiSARD system is a data-based method, and it is therefore used to detect and diagnose faults that have occurred in the past. Future works may improve the proposed model for evaluating, detecting and classifying new, unknown or multiple failures in the process.

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